25901448 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Skill Enhancement Course (SEC) - MG4SECPHY200 - ELECTRICAL CIRCUITS AND NETWORK SKILLS

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer all questions

Each question carries **1** marks

1. Electrical energy consumed is given by [A] / [CO3]

   a). V × I × Time

   b). V × I × Power factor × Time

   c). V × Power factor × Time

   d). V × I × Power factor

2. In AC circuits, current is opposed by [A] / [CO3]

   a). Only resistance

   b). Resistance and Inductance

   c). Resistance, Inductance and Capacitance

d). Only Capacitance

3. Kirchhoff's Voltage Law applies to which of the following? [U] / [CO3]

a).An open wire b).A closed-circuit loop

c).A single node d).A capacitor only

4. A galvanometer can measure only small currents. To use it as an ammeter, it must be [A] / [CO1]

a). Connected in parallel without modification

b). Used with a high resistance in series

c). Provided with a low resistance shunt in parallel

d). Used with a capacitor

5. What is the resultant resistance when two 2 Ω resistors are connected in series? [U] / [CO1]

a). 1 Ω

b). 2 Ω

c). 4 Ω

d). None of these

6. The part of a transformer that carries magnetic flux is the: [U] / [CO4]

a). Terminal

b). Winding

c). Core

d). Insulator

7. In a series circuit, the current is [K] / [CO4]

a).Different in each component

b).Zero

c).Maximum

d).Same through all components

8. The difference between synchronous speed and rotor speed is called [K] / [CO4]

a).Efficiency

b).Torque

c).Power factor

d).Slip

9. Surge is a [K] / [CO6]

a). Long voltage drop

b). Short voltage increase

c). constant Voltage d). Constant Current

10. In delta connection, the wires are connected [K] / [CO6]

a). In parallel b). To neutral

c). End to end d). To earth

[1x10 = 10]

Part B

Short Answer

Answer any **8** questions

Each question carries **5** marks

11. What is the physical meaning of real power in household appliances? [U] / [CO3]

12. Distinguish between Series and parallel connection. [U] / [CO3]

13. Match with the best option. [U] / [CO1]

(A) Charge	(i) Energy
(B) potential difference	(ii) temperature
(C) power	(iii) current
(D) volt x ampere	(iv) voltage
(E) resistance	(v) watt

14. Distinguish between single-phase and three-phase AC sources [U] / [CO3]

15. Explain the response of a capacitor when connected to DC and AC sources. [U] / [CO5]

16. Describe how the speed of a DC motor can be controlled. [U] / [CO4]

17. What is inductance? Describe self-inductance and give the formula for inductive reactance. [U] / [CO4]

18. Explain the working principles of following Disconnect switches
a) knife switch b) Rotary switch c) isolator [U] / [CO6]

19. Describe the colour code of wires used in an extension board and explain the significance of each wire. [A] / [CO6]

20. Write a note on the role of insulators in cables. [A] / [CO6]

[5x8 = 40]